

MELVIN JAN MARTINEZ

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EXPERIENCE

Cyber Threat Defense Engineer

TrendAi

August 2025 – Present

Metro Manila, Philippines

- Completed intensive malware analysis training rotating across web, email, and TCAT security teams, defending findings on assigned malware samples through daily exercises and formal panel assessments, then advanced into Team NVRT after passing midterm and final evaluations.
- Engineer detection patterns and filter rules for Deep Discovery Inspector (DDI) referencing CVE data, while handling up to 10 case investigations weekly — analyzing logs and PCAPs to resolve false positives/negatives and modifying patterns as needed.
- Built a unified AI-powered platform consolidating DDI pattern creation, case handling, VulDB, and CVE lookups into one dashboard, eliminating the need to cross-reference multiple separate tools, with AI-assisted triage that suggests case classification and relevant patterns to watch for log-inquiry cases.

Data Developer Support (Pioneer Batch)

Taisei Electronics Philippines, Inc.

January 2025 – April 2025

LIMA Estate, Lipa City

- Built database, backend, and frontend components following SDLC methodology as part of the pioneer team on a unified company system, replacing 4-5 fragmented legacy tools used across Sales Admin and HR.
- Conducted stakeholder interviews with Sales Admin and HR teams to scope automation needs, translating requirements into automated data entry and reporting features that reduced manual processing time by 70%.
- Integrated previously siloed Sales Admin and HR systems into a single platform, improving data accuracy and cross-department visibility.

PROJECTS

POS System – "Pause" (Grocery)

- Built a point-of-sale system using Python, HTML, CSS, and JavaScript, managing 7,000+ inputted products and retaining up to 1 year of monthly transaction history for reporting and audits.
- Implemented QR-based admin authentication for cash handling and void transactions, modernizing manual approval processes and reducing authorization errors.

ESURE – Electric Scooter Utilization with Rental System and Enhanced Safety (Capstone)

- Developed dual owner-and-customer Android applications in Android Studio for an e-scooter rental system, integrating GPS tracking refreshed every 10 seconds and immediate accident-detection alerts for rider safety.
- Engineered a simulated third-party payment gateway to replicate real-time transaction processing (GCash integration planned for production), enabling end-to-end booking-to-payment flow within the capstone environment.
- Designed the embedded controller prototype, docking/charging concept, and trip-suggestion algorithm for the e-scooter hardware, integrating microcontroller-level systems with the mobile app to improve accessibility and safety in the community.

Smart Door Lock System (ESP32, Android Studio)

- Built an IoT smart lock system using ESP32, solenoid, and relay hardware paired with a companion Android app, enabling one-tap remote unlocking.
- Designed an away-mode toggle that triggers immediate push notifications on suspected break-in attempts, adding a real-time security layer beyond standard lock systems.

EDUCATION

Bachelor of Science in Computer Engineering

STI College Lipa

2021 – 2025

Lipa City, Batangas

- Concentrated in robotics and embedded systems, developing hardware-level programming skills using ESP32 and Arduino microcontrollers across coursework projects.
- Led capstone team in designing a first-of-its-kind e-scooter rental system for the local community, earning multiple first-place exhibit awards for innovation and execution (full technical build detailed under Projects – ESURE).

Senior High School, STEM Strand

The Mabini Academy

2018 – 2020

Lipa City, Batangas

- Centralized coursework in Science and Technology, building foundational research, problem-solving, and analytical skills applied later in engineering studies.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, C++, C#, C, HTML, CSS, XML

Security / Analysis: PCAP Analysis, Log Analysis, Detection & Correlation Rule Tuning, CVE Referencing, Pattern Creation

Tools / Platforms: Deep Discovery Inspector (DDI), AWS (Cloud Foundations), Android Studio

Mobile / IoT: Android App Development, ESP32, Arduino, Embedded Systems, Microcontroller Programming

AI / Automation: AI Prompt Engineering, AI-Assisted Case Triage, Workflow Automation

LEADERSHIP & AFFILIATIONS

- Committee on Altar Servers – Mary Mediatrix of All Grace Parish | President (2016–2020), Member (2013–2023)
- Computer Engineers for Advancing Technology Society | 3rd Year Representative (2023)
- TMA Band | Treasurer (2019–2020), Member (2018–2020)

AWARDS

- Leadership Award (2025)
- College Research Exhibit (2024): Best in Overall Exhibit, Best in Communication, and Best in Usage of College Theme — First Place; Best in Implementation — Second Place

CERTIFICATIONS

- AWS Academy Graduate – AWS Academy Cloud Foundations (November 2024)
- Java Foundations – Oracle Academy (June 2022)
- EBA Scholar (2nd – 4th Year)